Networking Fundamentals for Systems Engineers

The TCP/IP Model

Most modern networking is built around the TCP/IP model, a layered framework that separates concerns so that engineers can reason about one layer without needing to understand every detail of the others. The link layer handles physical transmission of bits over a medium, the internet layer handles addressing and routing between networks using IP addresses, the transport layer manages end-to-end communication using protocols like TCP and UDP, and the application layer defines protocols that applications actually use, such as HTTP or DNS.

TCP versus UDP

TCP is a connection-oriented protocol that guarantees reliable, ordered delivery of data by using acknowledgments, retransmission, and flow control, making it the right choice for things like file transfers or web pages where correctness matters more than raw speed. UDP, by contrast, is connectionless and offers no delivery guarantees, but it has much lower overhead, which makes it preferable for real-time applications such as video calls or online games, where a dropped packet is less costly than the delay of waiting for retransmission.

Virtual Private Networks

A VPN creates an encrypted tunnel between two endpoints over an otherwise untrusted network, such as the public internet, allowing traffic to travel as if the endpoints were on the same private network. WireGuard has become popular as a lightweight, modern VPN protocol because its codebase is dramatically smaller than older protocols like OpenIPsec or OpenVPN, which makes it easier to audit for security issues while also achieving better throughput and faster connection setup on typical hardware.

Subnetting Basics

Subnetting divides a larger network into smaller, logically separate segments, which improves both security and traffic management. A subnet mask determines which portion of an IP address identifies the network and which portion identifies a specific host within it. Classless Inter-Domain Routing notation, such as writing an address as 192.168.1.0/24, expresses this division compactly and has largely replaced the older, less flexible system of fixed address classes.